

Highlights

Does AI rate through the cycle? Procyclicality in credit assessment

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- First controlled test of procyclicality in LLM credit ratings.
- All five production-tier LLMs downgrade identical firms as the macro worsens.
- Pooled effect: 0.43 rating notches of downgrade per macro-severity step.
- The effect is asymmetric: about 3x steeper into recessions than booms.
- A “through-the-cycle” prompt anchors ratings rather than de-cyclifying them.

Does AI rate through the cycle? Procyclicality in credit assessment

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ABSTRACT

Large language models (LLMs) are increasingly proposed as credit-rating and default-assessment tools, yet whether they rate “through the cycle” or amplify it—a property prudential regulators care about but that has been asserted rather than measured—has remained untested. We hold synthetic firm fundamentals byte-identical across an ordered five-point macroeconomic-severity axis (plus a matched credit-irrelevant placebo axis) and elicit letter ratings and one-year default probabilities from five production-tier models (two families, two capability tiers), so that any rating movement is procyclicality by construction, identified without the unobserved-fundamentals confound that limits the classical Amato–Furfine test. Across 32,000 ratings, every model is significantly procyclical (pooled 0.429 rating notches of downgrade per severity step, 95% CI [0.41, 0.45]; a 1.76-notch boom-to-recession swing), the effect is sharply asymmetric (3.2× steeper into recessions than out of them), an order of magnitude larger than the placebo, and robust to model-version fixed effects and a token-logprobability estimator. An explicit “through-the-cycle” instruction lowers the rating slope but does not restore through-the-cycle behaviour—it anchors the rating while suppressing the point-in-time default probability that should move, whereas a naive “be stable” instruction preserves that distinction better. Because model choice and prompt framing are first-order determinants of the effect, LLM-based credit assessment can silently import the capital-procyclicality channel that regulation seeks to dampen.

1. Introduction

Credit ratings shape regulatory capital, investment mandates, and the price of debt. A long-standing prudential concern is that ratings are procyclical—downgraded in downturns and upgraded in booms even when a firm’s fundamentals have not changed—because such movements amplify the credit cycle and, through internal-ratings-based (IRB) capital rules, tighten lending precisely when the economy is weakest (Kashyap and Stein, 2004; Gordy and Howells, 2006). Rating agencies aim to mitigate this by rating “through the cycle” (TTC): looking through transitory macroeconomic conditions to a firm’s creditworthiness over a full cycle (Amato and Furfine, 2004; Löffler, 2004), though whether agencies achieve it is itself debated (Löffler, 2013).

Large language models are now being deployed for credit assessment, and supervisors have flagged that generative models could re-introduce procyclicality (Aldasoro, Gambacorta, Korinek, Shreeti and Stein, 2024). This is plausible because LLMs behave as context-sensitive economic agents (Horton, 2023; Binz and Schulz, 2023) and, like human decision-makers, are susceptible to framing (Tversky and Kahneman, 1981); a shared model could also homogenise outcomes across users (Bommasani, Creel, Bansal, Guha and Liang, 2022). The concern, however, has been asserted rather than measured: it is not known whether LLM ratings move with the macroeconomy on fixed fundamentals, by how much, or whether the standard mitigation—instructing the model to rate through the cycle—works. Measuring this for human agencies is hard because fundamentals and the cycle move together, so the classical ordered-probit test of Amato and Furfine (2004) must proxy for unobserved fundamentals. Recent evidence that frontier LLMs reproduce date-specific human economic sentiment (Graham, Harvey and Jha, 2026) makes the question timely and the contamination risk concrete.

We provide the first controlled measurement of rating procyclicality in LLMs. Our design removes the identification problem by construction: a synthetic firm’s fundamentals are held byte-identical across an

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ordered macroeconomic-severity axis, so any change in the model’s rating is procyclicality, not a response to changed fundamentals—an experimental counterpart to the observational Amato–Furfine test. The benchmark is not that any movement is bad: a genuine through-the-cycle rater should hold the *rating* steady while its point-in-time default probability still tracks conditions, and we test that joint condition directly. We make three contributions. First, we show that all five production-tier models tested (spanning two families and two capability tiers) downgrade identical firms as the described macroeconomy worsens, by economically meaningful and statistically significant margins, and that model identity moves the magnitude roughly threefold. Second, we document that the effect is strongly *asymmetric*—about three times steeper into recessions than out of them—the precise pattern that transmits rating-driven capital procyclicality. Third, using a matched placebo instruction and the models’ default-probability outputs, we show that the obvious remedy—telling the model to rate “through the cycle”—does not restore through-the-cycle behaviour but instead induces generic *anchoring* (Tversky and Kahneman, 1974), suppressing the point-in-time default probability that should move. The paper proceeds as follows. Section 2 positions the contribution; Section 3 sets out the design and identification; Section 4 reports the results; Section 5 concludes.

2. Related literature

Our study connects three strands. The first is the long literature on *rating procyclicality* and through-the-cycle methodology. Amato and Furfine (2004) formalise the ordered-probit test for whether ratings move with the cycle conditional on fundamentals, and Löffler (2004, 2013) characterise—and question—the extent to which agencies rate through the cycle; the prudential stakes are set by the interaction of ratings with Basel capital rules (Kashyap and Stein, 2004; Gordy and Howells, 2006), with sovereign and corporate ratings alike carrying real pricing consequences (Cantor and Packer, 1996). Our experimental design sidesteps these studies’ central difficulty—that fundamentals and the cycle are observationally entangled—by holding fundamentals fixed. The second strand is the rapid adoption of *LLMs in finance*, for which this journal has become a leading outlet (Ali, Zafar and Aysan, 2025). Language models are used for return prediction (Lopez-Lira and Tang, 2023), financial-statement analysis (Kim, Muhn and Nikolaev, 2024), financial advice (Fieberg, Hornuf, Meiler and Streich, 2025; Kirschbaum, Keiper, Mölders and Zülch, 2026), and as simulated economic agents (Horton, 2023); within finance specifically they have been turned to measuring monetary-policy uncertainty (Ito, Sato and Ota, 2025), assessing the financial conditions of AI firms (Giaretta and Zen, 2026), and—most relevant here—to reliability evidence in banking supervision (García-Llorente and Olmeda, 2026). A parallel line warns that machine-generated scores and disclosures can mislead: greenwashing in ESG scores (Kathan, Utz, Dorfleitner, Eckberg and Chmel, 2025) and “AI washing” in corporate disclosure (Song, Hou, Ouyang and Hao, 2026). Broader analyses consider AI’s systemic and macroeconomic footprint, from agent-based market (in)stability (Llacay and Pfeffer, 2026) to the skill premium (Bloom, Prettnner, Saadaoui and Veruete, 2025), while supervisors have flagged the potential to re-introduce procyclicality (Aldasoro et al., 2024). Closest to us, Hens and Nordlie (2025) show in this journal that LLM-generated client risk profiles are statistically indistinguishable from those of experienced bankers in level, while the models’ explanations are generic—level-agreement without genuine reasoning. Our result is the troubling complement: on the credit-rating task, LLMs not only match a level but *move* it with context that a through-the-cycle rater should ignore. The third strand is *behavioural and cognitive limitations of LLMs*: models reproduce classic human framing and anchoring effects (Tversky and Kahneman, 1974, 1981; Binz and Schulz, 2023), are imperfectly calibrated (Kadavath, Conerly, Askill et al., 2022), and—through shared foundation models—can homogenise decisions across users (Bommasani et al., 2022). Procyclicality on fixed fundamentals is exactly such a context-driven bias, made consequential by its regulatory setting.

3. Design and identification

The unit of analysis is a rating cell = (model, prompt framing, seed, firm, macro state). We generate 80 synthetic firms with fixed fundamentals (revenue growth, EBITDA margin, leverage, interest coverage, free cash flow, liquidity, and Altman-style ratios; Altman, 1968) stratified across five credit-quality tiers and eight sectors, so the fundamentals-implied ratings span the scale (AAA–CCC). Each firm’s fundamentals are serialised once and reused byte-identically under every macro state; only a prepended macro-environment

block changes. The macro axis has five ordered levels—boom, expansion, neutral, slowdown, severe recession (severity $-2 \dots +2$)—each carried by matched, number-based indicators (GDP growth, unemployment, credit spreads, default rate, lending standards) with no calendar dates, so a model cannot pattern-match a real historical episode.

Two features sharpen identification. First, a matched-format but credit-irrelevant placebo axis (regional weather and traffic indicators, same structure and number density) measures pure frame-compliance; the net (macro minus placebo) coefficient is the demand-effect-adjusted cyclicity. Second, the per-cell random seed is keyed on the firm and framing but omits the macro state, so a firm receives an identical seed across all states—sampling noise cannot masquerade as a cycle effect. Each model returns a letter rating (mapped to an S&P notch, 1–21) and a one-year probability of default (PD).

We study four prompt framings—terse, point-in-time, explicit through-the-cycle, and a “be stable” instruction with no cycle reasoning—on five models: three OpenAI (`gpt-4o-mini`, `gpt-4.1-mini`, `gpt-4o`) and two Google Gemini Flash-tier models accessed via the `gemini-flash-latest` and `gemini-flash-lite-latest` aliases, all queried in July 2026 with the served-model snapshot/`system_fingerprint` logged per call for reproducibility. We use two seeds at temperature zero. The primary estimand is the mean within-firm slope of rating notch on macro severity (notches of downgrade per +1 step; positive = procyclical), with firm-clustered (tier-stratified) bootstrap 95% confidence intervals (2,000 draws). We also report the boom-to-recession swing, downside vs upside half-slopes, the Amato–Furfine ordered-probit coefficient, and—for the OpenAI models—a continuous expected-notch slope from the rating-token log-probabilities. The full run comprises 32,000 ratings for US\$13.69, with zero parse errors; every capture is hashed and the analysis is a deterministic function of the frozen data.

4. Results

4.1. LLMs are procyclical on identical fundamentals

On byte-identical fundamentals, every model downgrades as the described macroeconomy worsens (Table 1; Fig. 1a). Pooling across models under the terse framing, ratings fall by 0.429 notches per severity step (95% CI [0.41, 0.45])—a 1.76-notch swing from boom to severe recession, enough to move a BBB firm toward BB+/BB with nothing real changing. The per-model coefficient ranges from 0.230 (Gemini Flash) to 0.775 (GPT-4o-mini), and every model’s CI excludes zero. Model identity is thus a first-order determinant—a roughly threefold spread—and, within OpenAI, the larger GPT-4o (0.387) is about half as procyclical as GPT-4o-mini.

The effect is not framing suggestibility: the matched credit-irrelevant placebo axis moves ratings only 0.042 notches per step, so the net coefficient is 0.387 (95% CI [0.36, 0.41]), still excluding zero. It is not infrastructure drift: adding served-model (system-fingerprint) fixed effects shifts the coefficient by less than 0.003. It is not a censoring or parsing artefact: every model parses at 1.000, with the largest-cyclicity model (GPT-4o-mini) among the cleanest. For the OpenAI models, a continuous expected-notch slope from rating-token log-probabilities reproduces the sampled estimate (e.g., GPT-4o-mini 0.775 vs 0.765), and the Amato–Furfine ordered-probit coefficient agrees in sign throughout and is of comparable magnitude, though the two estimators do not perfectly preserve the cross-model rank order (Appendix Table 3).

The magnitudes are economically material: a pooled 1.76-notch swing means that a firm rated, say, A— in a boom is rated near BBB in a severe recession on unchanged fundamentals, and for the most sensitive model the swing exceeds three notches; we translate this into a capital requirement in §4.4. That the coefficient survives the placebo control, the log-probability estimator, served-model fixed effects and the ordered-probit null makes it unlikely to be an artefact of suggestibility, sampling noise or infrastructure drift.

4.2. The procyclicality is asymmetric

Splitting the macro axis at neutral, ratings fall much faster going into recession than they rise going into a boom. Pooled, the downside half-slope (neutral to severe recession) is 0.67 (95% CI [0.63, 0.72]) versus an upside half-slope (boom to neutral) of 0.21 (95% CI [0.18, 0.24])—a $3.2\times$ asymmetry that holds for every model (Fig. 2). This is the financially dangerous margin: downturn downgrades dominate, exactly the pattern that transmits rating-driven capital procyclicality, and it echoes the stricter-in-downturns behaviour long attributed to human agencies (Löfller, 2004). Because capital requirements are convex in the rating and bind

hardest in stress, an asymmetric downgrade profile is more procyclical than a symmetric one of equal average slope, amplifying the prudential concern.

4.3. “Through-the-cycle” instructions anchor rather than de-cyclify

Prompt framing moves the coefficient monotonically (Table 2; Fig. 1b): point-in-time 0.516, terse 0.429, “be stable” 0.329, and through-the-cycle 0.160 (95% CI [0.14, 0.18]); the terse-minus-TTC contrast is 0.269 notches per step. Taken alone, this looks like a successful mitigation. It is not. We define genuine through-the-cycle behaviour as a joint condition: the rating slope should fall toward zero *and* the (point-in-time) default probability should keep moving with the cycle. Anchoring collapses both. We measure the default probability on the log-odds scale (so the comparison across framings is not driven by level differences, and is robust to the miscalibration of verbalised LLM probabilities; Kadavath et al., 2022; Kang and Liu, 2023). Pooled, the log-odds PD slope falls from 0.270 under terse to 0.102 under the through-the-cycle instruction—a *retention* of only 0.38 (95% CI [0.35, 0.40]), far below the value near one that genuine through-the-cycle behaviour requires; the suppression is comparable on the downside and upside (retention 0.35 and 0.38). So the model satisfies the first condition and violates the second: it stabilises the wrong object. Revealingly, the vaguer “be stable” instruction, which is not semantically about the cycle, retains more PD movement while still calming the rating. A residual downgrade survives everywhere—up to 1.60 notches across the cycle for GPT-4o-mini.

4.4. Capital consequences

The material question is not merely that the rating moves but what the movement costs. We translate the rating path into capital by anchoring through the *rating*—which is what enters a credit pipeline—rather than the disclaimed verbalised PD: we map each notch to a stylised one-year default probability calibrated to published agency average default rates, and apply the Basel internal-ratings-based (IRB) corporate risk-weight formula. This is a stylised proxy for the core economic logic (loss given default 0.45, maturity 2.5, the corporate asset-correlation function), not the regulatory framework in full detail (no SME or firm-size adjustment). Reporting the *change* across the severity axis, an identical firm’s required capital rises by a pooled 1.87 percentage points of exposure from boom to severe recession (95% CI [1.73, 2.00]), and by 3.48 pp for the most sensitive model. The verbalised-PD variant agrees in direction (2.92 pp). Because the risk weight is convex in PD, the §4.2 asymmetry compounds: the downside capital increase (neutral→recession, 1.55 pp) is about five times the upside decrease (boom→neutral, 0.32 pp)—a wider gap than the notch asymmetry alone, so an equal-average-slope symmetric rater would produce a smaller capital swing than we observe.

5. Conclusion

Using a design in which fundamentals are fixed by construction, we measure rating procyclicality in large language models directly. All five models downgrade identical firms as the macroeconomy is described as worsening, by economically meaningful and statistically robust margins, and they do so asymmetrically—punishing simulated downturns far more than they reward simulated expansions. The magnitude is a first-order function of model identity, so in a deployed pipeline the choice of model is itself a prudential decision. The natural mitigation—instructing the model to rate through the cycle—fails instructively: rather than holding the rating steady while letting the default probability track conditions, it suppresses both, and it leaves a measurable residual. Our design perturbs a macro block inside the prompt; this is representative rather than artificial, because realistic credit pipelines routinely place macroeconomic context in front of the model—through retrieval over news and filings, analyst preambles, or market-data conditioning—so the sensitivity we measure is precisely the exposure a deployment inherits. These results should be read against their scope—a synthetic firm battery built for identification rather than external realism, five production-tier models from two families, and prompt-level rather than fine-tuning interventions—and they motivate benchmarking LLM procyclicality against agency and IRB baselines, testing fine-tuning and retrieval remedies that target the point-in-time object directly, and, in the interim, treating model selection and prompt framing as governed parameters wherever LLMs touch credit and capital decisions.

Table 1

Per-model rating procyclicality on byte-identical fundamentals. β = notches of downgrade per +1 macro-severity step (terse framing); firm-clustered bootstrap 95% CIs. Swing = boom→severe-recession notches. Asymmetry = downside (into recession) vs upside (into boom) half-slopes. Residual = β after the through-the-cycle instruction; PD-ret. = PD-slope retention under that instruction.

Model	β	95% CI	Swing	Down	Up	D:U	Resid.	PD-ret.
Gemini Flash	0.230	[0.19, 0.27]	0.96	0.38	0.10	4.0	0.059	0.22
Gemini Flash-Lite	0.285	[0.25, 0.32]	1.15	0.45	0.12	3.6	0.056	0.34
GPT-4.1-mini	0.471	[0.41, 0.53]	1.94	0.77	0.20	3.9	0.096	0.24
GPT-4o	0.387	[0.34, 0.43]	1.55	0.59	0.18	3.3	0.187	0.52
GPT-4o-mini	0.775	[0.69, 0.86]	3.18	1.15	0.44	2.6	0.402	0.59

Table 2

Pooled procyclicality by prompt framing (across five models), notches per severity step with firm-clustered bootstrap 95% CIs.

Framing	Pooled β	95% CI	Excl. 0
Point-in-time	0.516	[0.49, 0.54]	True
Terse	0.429	[0.41, 0.45]	True
"Be stable"	0.329	[0.30, 0.35]	True
Through-the-cycle	0.160	[0.14, 0.18]	True

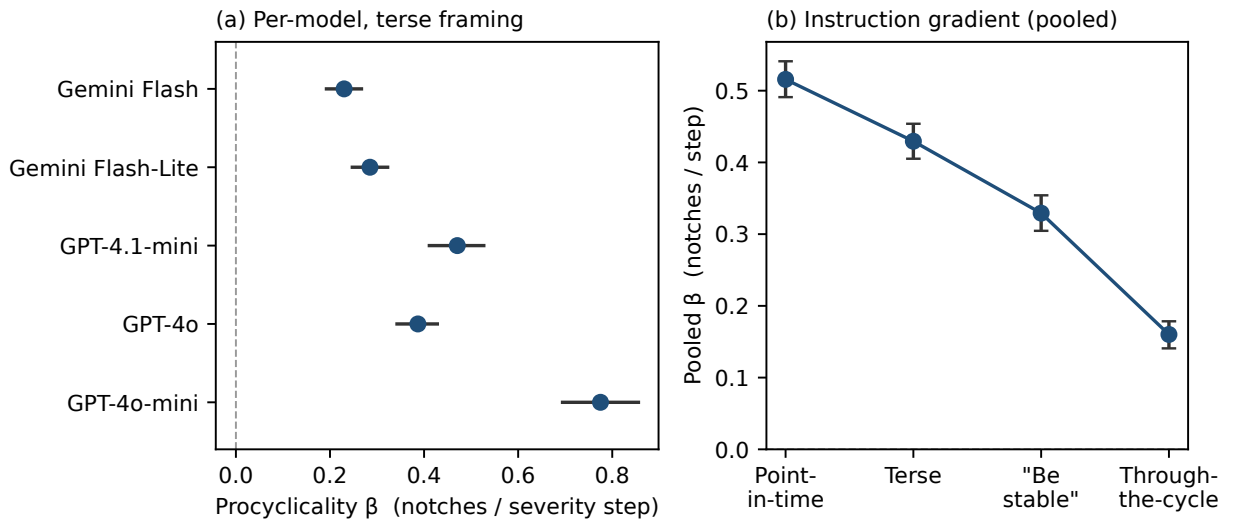


Figure 1: (a) Per-model procyclicality β under the terse framing with 95% CIs; the dashed line marks the through-the-cycle null ($\beta = 0$). (b) Pooled β across the four prompt framings (point-in-time → terse → "be stable" → through-the-cycle) with 95% CIs.

Declaration of generative AI use

During the preparation of this work the authors used AI-assisted coding and drafting tools to help implement the data pipeline and draft the manuscript. After using these tools the authors reviewed and edited the content and take full responsibility for the content of the published article. The large language models studied are the object of the research, not tools of manuscript preparation.

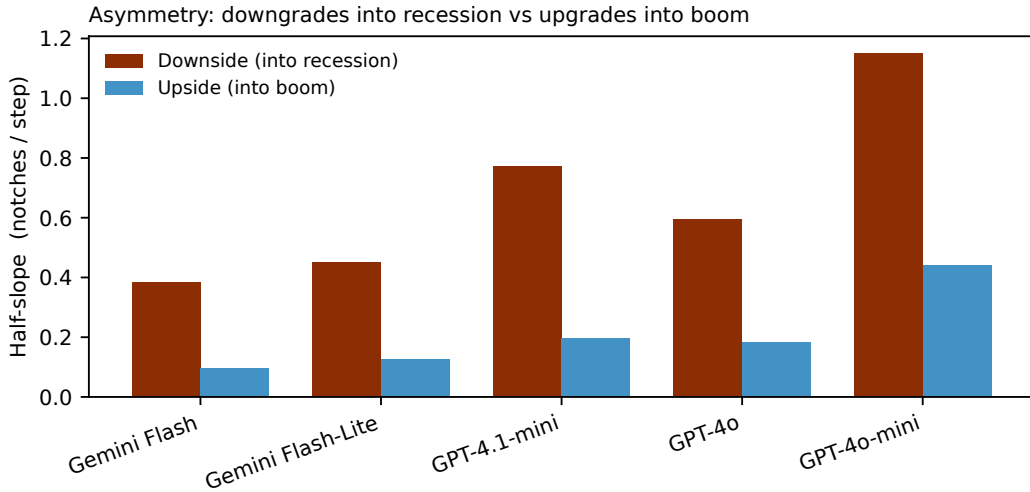


Figure 2: Asymmetry: per-model downside half-slope (neutral→severe recession) versus upside half-slope (boom→neutral), terse framing. Every model downgrades far faster into recessions than it upgrades into booms.

Table 3

Robustness of the terse-framing coefficient: placebo-adjusted (net), log-probability expected-notch (OpenAI only), within-cell seed noise floor, and the Amato–Furfine ordered-probit estimate.

Model	β	Net	Exp-notch	Noise SD	Probit
Gemini Flash	0.230	0.190		0.020	0.272
Gemini Flash-Lite	0.285	0.268		0.000	0.235
GPT-4.1-mini	0.471	0.464	0.420	0.086	0.307
GPT-4o	0.387	0.367	0.378	0.025	0.403
GPT-4o-mini	0.775	0.646	0.765	0.072	0.726

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All data and code are openly available at <https://github.com/samirrc2/Does-AI-rate-through-the-cycle---> a citable Zenodo archive accompanies the published version. Reproduction is offline, deterministic and free (Chincholikar and Chawla, 2026).

A. Robustness

Adding served-model (`system_fingerprint`) fixed effects shifts the terse coefficient by < 0.003 for every OpenAI model; a within-cell seed noise floor confirms the signal exceeds residual non-determinism. The ordered-probit (Amato–Furfine) null uses the firm mean notch as a Mundlak control. The capital translation (§4.4) maps each notch to a stylised one-year default probability (AAA $\approx 0.01\%$, BBB $\approx 0.25\%$, BB $\approx 1.1\%$,

$B \approx 5\%$, $CCC \approx 20\%$; monotone across the 21-notch scale, calibrated to published agency averages) and applies the Basel IRB corporate formula; the reported *change* in capital is robust to the exact mapping.

B. Prompts

The prompt framings and the macro and placebo blocks below are extracted programmatically from the capture code (`agent.py`) and reproduce the executed instrument verbatim.

B.1. Prompt framings

The four instruction framings (system prompt), verbatim as executed:

v_terse

You are a credit rating analyst. Assign a long-term issuer credit rating to the company described

v_ttc

You are a credit rating analyst applying a THROUGH-THE-CYCLE methodology. Assign a long-term issuer

v_pit

You are a credit rating analyst applying a POINT-IN-TIME methodology. Assign a long-term issuer

v_stable

You are a credit rating analyst. Assign a long-term issuer credit rating that is STABLE and does

B.2. Macro-environment blocks

Prepended to the byte-identical fundamentals; only these five blocks differ across the severity axis:

boom

MACRO ENVIRONMENT (current conditions, this fiscal period):

- Real GDP growth (annualized): +4.5%
- Unemployment rate: 3.4%
- Investment-grade corporate credit spread: 95 bps
- Trailing 12-month corporate default rate: 0.6%
- Bank lending standards: easing broadly

expansion

MACRO ENVIRONMENT (current conditions, this fiscal period):

- Real GDP growth (annualized): +2.6%
- Unemployment rate: 4.2%
- Investment-grade corporate credit spread: 135 bps
- Trailing 12-month corporate default rate: 1.4%
- Bank lending standards: modestly easing

neutral

MACRO ENVIRONMENT (current conditions, this fiscal period):

- Real GDP growth (annualized): +1.8%
- Unemployment rate: 5.0%
- Investment-grade corporate credit spread: 175 bps
- Trailing 12-month corporate default rate: 2.3%
- Bank lending standards: unchanged

slowdown

MACRO ENVIRONMENT (current conditions, this fiscal period):

- Real GDP growth (annualized): -0.8%
- Unemployment rate: 6.4%
- Investment-grade corporate credit spread: 320 bps
- Trailing 12-month corporate default rate: 4.1%
- Bank lending standards: tightening

severe_recession

MACRO ENVIRONMENT (current conditions, this fiscal period):

- Real GDP growth (annualized): -3.6%
- Unemployment rate: 8.9%
- Investment-grade corporate credit spread: 620 bps
- Trailing 12-month corporate default rate: 7.5%
- Bank lending standards: sharply tightening

B.3. Placebo blocks (credit-irrelevant control)

Matched in structure and number density, on a credit-irrelevant axis:

placebo_calm

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: +0.6 degC
- Daylight hours vs annual mean: +4.5%
- Monthly rainfall vs normal: 95 mm
- Regional air-quality index: 38
- Commuter traffic congestion: light

placebo_mild

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: +0.2 degC
- Daylight hours vs annual mean: +2.6%
- Monthly rainfall vs normal: 135 mm
- Regional air-quality index: 52
- Commuter traffic congestion: moderate

placebo_normal

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: 0.0 degC
- Daylight hours vs annual mean: +1.8%
- Monthly rainfall vs normal: 175 mm
- Regional air-quality index: 60
- Commuter traffic congestion: typical

placebo_rough

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: -0.4 degC
- Daylight hours vs annual mean: -0.8%
- Monthly rainfall vs normal: 320 mm
- Regional air-quality index: 88
- Commuter traffic congestion: heavy

placebo_storm

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: -1.1 degC
- Daylight hours vs annual mean: -3.6%
- Monthly rainfall vs normal: 620 mm
- Regional air-quality index: 140
- Commuter traffic congestion: severe

CRediT authorship contribution statement

Samir Chincholikar: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Robin Chawla:** Conceptualization, Validation, Writing – review and editing, Supervision.

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